Adam Lechowicz

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adamlechowicz.github.io

Education

University of Massachusetts Amherst (Manning CICS), *Ph.D. Computer Science* 2022 –
Supported by DOE Computational Science Graduate Fellowship, 2023 –
Advised by: Mohammad Hajiesmaili and Prashant Shenoy

University of Massachusetts Amherst 2019 – 2022
B.S. Computer Science & B.A. Political Science, summa cum laude
Phi Beta Kappa, 21st Century Leader

Research Interests

I work towards a future in which the infrastructure behind everyday life, from data centers to energy systems, continually adapts to reduce its environmental impact while delivering dependable services. I develop systems and algorithms that expose, enable, and exploit this flexibility. A core thread of my work is learning-augmented algorithms, which pair the performance of AI with worst-case guarantees, so decisions improve when predictions are accurate and stay dependable when they are not.

Experience

Research Intern **Redmond, Wash.**
Systems and Networking Group, Microsoft Research Summer 2026
- Mentored by Behnaz Arzani, Pooria Namyar, and Konstantina Mellou

Ph.D. Intern **Golden, Colo.**
Power Systems Engineering Center, National Renewable Energy Laboratory (now NLR) Summer 2024
- Advised by Joshua Comden and Andrey Bernstein

Visiting Researcher **Pasadena, Calif.**
Computing and Mathematical Sciences, California Institute of Technology Summer 2023
- Advised by Adam Wierman

Research Assistant **Amherst, Mass.**
Manning College of Information and Computer Sciences Jun. 2022 – May 2023
- Advised by Mohammad Hajiesmaili and Prashant Shenoy

Awards and Honors

Georgia Tech, UC Berkeley, U. of Michigan: 2026 NextProf Nexus Workshop (*invited*) Fall 2026
UMass Graduate School: NSF Leadership Fellowship (*$2,000*) Fall 2025
Manning CICS: Ph.D. Portfolio with Distinction Fall 2024
Manning CICS: Dr. C. Mohan Graduate Scholarship (*$10,000*) Fall 2023
U. of Michigan, Georgia Tech, UC San Diego: 2023 NextProf Pathfinder Workshop (*invited*) Fall 2023
U.S. Department of Energy: Computational Science Graduate Fellowship (*$195,000*) 2023-2027
Energy Transition Institute: NSF ELEVATE Fellowship (*$38,000*) 2022-2024
Manning CICS: Outstanding Course Assistant Award Fall 2022
UMass Amherst: Class of 2022 – 21st Century Leader Award (*highest undergraduate honor*) Spring 2022
Manning CICS: Outstanding Undergraduate Achievement Award Spring 2022
UMass Amherst: 2021-2022 UMass Amherst Rising Researcher Spring 2022
UMass Amherst: Phi Beta Kappa Spring 2021

Publications

† – *indicates undergraduate I advised* $(\alpha-\beta)$ – *indicates alphabetical author order*

Conference Publications

Mahsa Sahebdel, Ali Zeynali, Adam Lechowicz, Bo Sun, and Mohammad Hajiesmaili
Online Joint Pricing and Scheduling in Distributed Networked Platforms
ACM Intl. Symp. on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks & Computing (MobiHoc), 2026.

Adam Lechowicz, Rohan Shenoy†, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, and Christina Delimitrou
♠ **SPADE: Signal-Aware DAG Scheduling and Dynamic Provisioning for Data Processing Clusters**
USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2026.

Adam Lechowicz, Nicolas Christianson, Mohammad Hajiesmaili, Adam Wierman, and Prashant Shenoy
Competitive and Learned Online Algorithms for HVAC Demand Management in Thermal Networks
ACM International Conference on Future Energy Systems (e-Energy), 2026.

Ezra Johnson†, Adam Lechowicz, and Mohammad Hajiesmaili
Signal-Aware Workload Shifting Algorithms with Uncertainty-Quantified Predictors
ACM International Conference on Future Energy Systems (e-Energy), 2026.

Cooper Sigrist, Anupama Sitaraman, Alice Zhang, Adam Lechowicz, Yuvraj Agarwal, and Mohammad Hajiesmaili
The Price to Site: Cost-Effective Incentives for Solar Deployment
ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), 2026.

Fabian Mkocheko, Walid A. Hanafy, Adam Lechowicz, David Irwin, and Prashant Shenoy
City-Scale Decarbonization of Residential Heating Under Transformer Constraints
ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), 2026.

Adam Lechowicz, Nicolas Christianson, Mohammad Hajiesmaili, Adam Wierman, and Prashant Shenoy
Online Smoothed Demand Management
ACM SIGMETRICS, 2026.

Cooper Sigrist, Archimedes Li, Pichsinee Lertsaroj, Ryan Boldi, Adam Lechowicz, Noman Bashir, and Mohammad Hajiesmaili
Multi-objective Evolutionary Learning for Near Pareto-Optimal Optimization of Solar Deployment ***(best paper candidate)***
ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), 2025.

Adam Lechowicz, Rohan Shenoy†, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, and Christina Delimitrou
Carbon- and Precedence-Aware Scheduling for Data Processing Clusters *(SIGCOMM short)*
ACM SIGCOMM, 2025.

Mohammadreza Daneshvaramoli, Helia Karisani, Adam Lechowicz, Bo Sun, Cameron Musco, and Mohammad Hajiesmaili
Near-Optimal Consistency-Robustness Trade-Offs for Learning-Augmented Online Knapsack Problems
International Conference on Machine Learning (ICML), 2025.

Adam Lechowicz, Joshua Comden, and Andrey Bernstein
Optimizing Individualized Incentives from Grid Measurements under Limited Knowledge of Agent Behavior
ACM International Conference on Future Energy Systems (e-Energy), 2025.

Cooper Sigrist, Adam Lechowicz, Jovan Champ, Noman Bashir, and Mohammad Hajiesmaili
Lost in Siting: The Hidden Carbon Cost of Inequitable Residential Solar Installations
ACM International Conference on Future Energy Systems (e-Energy), 2025.

Julia Köhlke, Adam Lechowicz, Oluwole Fabikun†, Noman Bashir, Abel Souza, Prashant Shenoy, and Sebastian Lehnhoff
Examining the Adoption of Electromobility Concepts Across Social Contexts for Energy Transition
ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), 2024.

Adam Lechowicz, Nicolas Christianson, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, and Prashant Shenoy
Chasing Convex Functions with Long-term Constraints
International Conference on Machine Learning (ICML), 2024.

Adam Lechowicz, Nicolas Christianson, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, and Prashant Shenoy
Online Conversion with Switching Costs: *Robust and Learning-augmented Algorithms*
ACM SIGMETRICS / IFIP Performance, 2024.

Roozbeh Bostandoost, Adam Lechowicz, Walid A. Hanafy, Noman Bashir, Prashant Shenoy, and Mohammad Hajiesmaili
LACS: Learning-Augmented Carbon-Aware Resource Scaling for Uncertain Demand
ACM International Conference on Future Energy Systems (e-Energy), 2024.

Adam Lechowicz, Rik Sengupta, Bo Sun, Shahin Kamali, and Mohammad Hajiesmaili
Time Fairness in Online Knapsack Problems
International Conference on Learning Representations (ICLR), 2024.

Adam Lechowicz, Noman Bashir, John Wamburu, Mohammad Hajiesmaili, and Prashant Shenoy
Equitable Network-Aware Decarbonization of Residential Heating at City Scale
ACM International Conference on Future Energy Systems (e-Energy), 2023.

Journal Publications

Nikita Bhalla, Adam Lechowicz, and Cameron Musco $(\alpha{-}\beta)$
Local Edge Dynamics and Opinion Polarization
ACM Transactions on Intelligent Systems and Technology, August 2025. Also in WSDM '23.

Anupama Sitaraman, Adam Lechowicz, Noman Bashir, Xutong Liu, Prashant Shenoy, and Mohammad Hajiesmaili
Dynamic Incentive Allocation for City-scale Deep Decarbonization
ACM Journal on Computing and Sustainable Societies, June 2025.

Adam Lechowicz, Nicolas Christianson, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, and Prashant Shenoy
Learning-Augmented Competitive Algorithms for Spatiotemporal Online Allocation with Deadline Constraints
Proc. of the ACM on Measurement and Analysis of Computing Systems 9(1), Mar. 2025. Also in SIGMETRICS '25.

Adam Lechowicz, Nicolas Christianson, Jinhang Zuo, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, and Prashant Shenoy
The Online Pause and Resume Problem: *Optimal Algorithms and An Application to Carbon-Aware Load Shifting*
Proc. of the ACM on Measurement and Analysis of Computing Systems 7(3), Dec. 2023. Also in SIGMETRICS / Performance '24.

Bhawana Chhaglani, Camellia Zakaria, Adam Lechowicz, Jeremy Gummeson, and Prashant Shenoy
FlowSense: Monitoring Airflow in Building Ventilation Systems Using Audio Sensing
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT), March 2022.

Refereed Workshop Papers

Roozbeh Bostandoost, Adam Lechowicz, Walid A. Hanafy, Prashant Shenoy, and Mohammad Hajiesmaili
Quantifying the Carbon Reduction of DAG Workloads: A Job Shop Scheduling Perspective
International Workshop on Systems and Methods for Sustainable Large-Scale AI (GreenSys), 2026.

Cooper Sigrist, Archimedes Li, Pichsinee Lertsaroj, Ryan Boldi, Adam Lechowicz, Noman Bashir, and Mohammad Hajiesmaili
Learning to Site: A Multi-objective Optimization of Rooftop Solar Installations using Evolutionary Neural Networks
ACM SIGMETRICS Performance Evaluation Review, August 2025.

Roozbeh Bostandoost, Walid A. Hanafy, Adam Lechowicz, Noman Bashir, Prashant Shenoy, and Mohammad Hajiesmaili
Data-driven Algorithm Selection for Carbon-Aware Scheduling
ACM SIGEnergy Energy Informatics Review 4(5), Dec. 2024. Also in HotCarbon '24.

Preprints and Papers Under Review

Helia Karisani, Mohammadreza Daneshvaramoli, Adam Lechowicz, Mingda Qiao, and Mohammad Hajiesmaili
Learning-Augmented Online Caching with Calibrated Phase Predictions
In submission, 2026.

Jackson Bibbens, Adam Lechowicz, Nicolas Christianson, Bo Sun, Mohammad Hajiesmaili
Peak-Aware One-Way Trading: Optimal Algorithms and Applications to Energy Procurement
In submission, 2026.

Adam Lechowicz, Nicolas Christianson, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, and Prashant Shenoy
Online Conversion with Switching Costs: *Robust and Learning-augmented Algorithms*
Journal version under review; Preliminary version appeared at ACM SIGMETRICS / IFIP Performance '24.

In Preparation

Adam Lechowicz, Nicolas Christianson, Mohammad Hajiesmaili, Adam Wierman, and Prashant Shenoy
Online Smoothed Demand Management
Journal version in preparation; Preliminary version accepted to ACM SIGMETRICS '26.

Adam Lechowicz, Pooria Namyar, Konstantina Mellou, Mohammad Hajiesmaili, Behnaz Arzani, and Dave Maltz
Concierge: Decision Support for Pricing and Admission Control of Large Capacity Requests
Conference paper in preparation.

Presentations

⋆ – indicates invited

Unlocking System Control Benefits of AI using Theoretical Modeling
Sustainable Compute & AI Infrastructure Research Symposium @ UMass Sep. 2026
⋆ ACM SIGMETRICS Young Researcher Lightning Talk Jun. 2026

SPADE: Signal-Aware DAG Scheduling and Dynamic Provisioning for Data Processing Clusters
USENIX OSDI 2026 Jul. 2026

Online Smoothed Demand Management
ACM SIGMETRICS 2026 Jun. 2026

Learning-Augmented Algorithms for Spatiotemporal Online Allocation with Deadline Constraints
⋆ *CS/OR/Econ Theory Lunch @ Yale* (Slides) Mar. 2026
⋆ *INFORMS Annual Meeting 2025* (Slides) Oct. 2025
ACM SIGMETRICS 2025 Jun. 2025

Carbon- and Precedence-Aware Scheduling for Data Processing Clusters
ACM SIGCOMM 2025 Sep. 2025

Learning-Augmented Online Optimization with Switching Costs and Long-term Constraints
⋆ *INFORMS Applied Probability Society Conference 2025* (Slides) Jul. 2025

Optimizing Individualized Incentives from Grid Measurements under Limited Knowledge of Agent Behavior
ACM e-Energy 2025 Jun. 2025
⋆ *RSRG / FALCON Group Meeting @ Caltech* (Slides) May 2025
ELEVATE Retreat, Energy Transition Institute @ UMass Jan. 2025
⋆ *ACM SIGEnergy Graduate Seminar* (Slides) Nov. 2024

Online Optimization with Switching Costs and Long-term Constraints
⋆ *SODALab Group Meeting @ University of Alberta* (Slides) Aug. 2024

Online Conversion with Switching Costs: *Robust and Learning-augmented Algorithms*
ACM SIGMETRICS / IFIP Performance 2024 Jun. 2024

The Online Pause and Resume Problem: *Optimal Algorithms and An Application to Carbon-Aware Load Shifting*
ACM SIGMETRICS / IFIP Performance 2024 Jun. 2024

Equitable Network-Aware Decarbonization of Residential Heating at City Scale
⋆ *PIT × UMass: Developing Tech for the Public Interest (**poster in library exhibit**)* Apr – May. 2024
ACM e-Energy 2023 Jun. 2023

Theoretical Foundations for Carbon-aware Load Shifting
ELEVATE Retreat, Energy Transition Institute @ UMass Feb. 2024

Fresh Challenges for Online Problems: new directions motivated by practice
⋆ *RSRG / FALCON Lunch Seminar @ Caltech* (Slides) Jun. 2023

Local Edge Dynamics and Opinion Polarization
ACM WSDM 2023 Feb. 2023

On the Necessary "Unfairness" of Competitive Online Algorithms
UMass CS Theory Seminar (Slides) Nov. 2022

Edge Dynamics and Opinion Polarization in Social Networks
Honors Thesis Defense (Slides) Apr. 2022

Machine Learning for Absolute Beginners – Mathematical Foundations of ML
Workshop @ HackUMass IX (Slides) Nov. 2021

Research Advising and Mentorship

Tri Than *(MS, UMass)* 2025 –
Urvi Gupta, Pihul Nyati, Samuel Nkemnkia, Rakshita Saroha – ERSP @ CICS 2025 – 2026
Ezra Johnson *(UG, Caltech)* – Computing for an Equitable Energy Transition REU Summer 2025
Rohan Shenoy *(UG, UC Berkeley)* – (*received NSF GRFP*) 2024 – 2026
Anisha Prathi *(UG, UMass)* – Undergraduate Honors Thesis 2023 – 2024
Kaosisochukwu Nwosu *(UG, UMass)* – (*received Outstanding Achievement Award*) 2023 – 2024
Lily Davoren *(UG, Bryn Mawr)* – Computing for an Equitable Energy Transition REU Summer 2023
Coo Katsuno, Saif Masoud, Riley Kim Connell – URV Program @ CICS Summer 2023
Oluwole Fabikun *(UG, UMass)* – LSAMP Scholar – (*featured in graduating "Senior Series"*) 2023 – 2024

Teaching

Guest Lecturer — **Amherst, Mass.**
INFO 203: A Networked World – College of Information and Computer Sciences — Spring 2026, Fall 2026
- Presented four lectures on the impact of data centers and related algorithmic challenges in an undergraduate Informatics course at UMass.

Lecturer & Mentor — **Amherst, Mass.**
UMass Turing Summer Program – College of Information and Computer Sciences — Summers 2022, 2023, 2025

Course Assistant — **Amherst, Mass.**
COMPSCI 389: Intro to Machine Learning – College of Information and Computer Sciences — Spring 2022

Service

– Program Committees –

Shadow PC – ACM SIGMETRICS 2026 — 2026

– Journal Reviewing –

IEEE Transactions on Parallel and Distributed Systems — 2026
IEEE Transactions on Industrial Informatics — 2026
Sustainable Computing: Informatics and Systems — 2026
Computers and Operations Research — 2026
ACM Transactions on Modeling and Performance Evaluation of Computing Systems — 2026
IEEE Transactions on Network Science and Engineering — 2025

– Conference Reviewing –

ACM Web Conference (WWW) — 2027
European Symposium on Algorithms (ESA) *(subreviewer)* — 2026
International Conference on Learning Representations (ICLR) — 2025, 2026
Conference on Neural Information Processing Systems (NeurIPS) — 2025, 2026
ACM SIGMETRICS *(subreviewer)* — 2024, 2025

– Workshop Organization & Reviewing –

AI Foundations for Power Grids (AI4PowerGrids) — NeurIPS '26
Workshop on Learning-augmented Algorithms: Theory and Applications (LATA) — SIGMETRICS '26
Workshop on Measurements, Modeling, & Metrics for Carbon-Aware Computing (CarbonMetrics) — SIGMETRICS '25, '26
Chair – Workshop on Learning-augmented Algorithms: Theory and Applications (LATA) — SIGMETRICS '25

– Department & University Service –

New Building Committee – Manning CICS — Jun. 2022 – ongoing
Ph.D. Grad Representative – Manning CICS — Sep. 2025 – Sep. 2026
Headmaster (Organizer) – OlympCICS – Manning CICS — Jan. 2024
- Organized 2nd annual college-wide competition for CS PhD students.

Grad Student Social Committee – Manning CICS — Sep. 2023 – Aug. 2024
New PhD Student Committee – Manning CICS — Sep. 2023 – Aug. 2024
Student Trustee – Board of Trustees, University of Massachusetts — Jul. 2022 – Jul. 2023
- Student representative of the Amherst campus on system-wide Board of Trustees.

Committee Against Racism and for Equity – Manning CICS — Sep. 2021 – Aug. 2022
Flexible Learning Task Force & Implementation Committee – UMass Amherst — Jan. 2021 – May 2022

– Outreach & Volunteering –

Judge – HackUMass XI & XIII (hackathon) — Nov. 2023, 2025
Workshop Co-lead and Presenter – Holyoke Energy Justice Leaders Workshop #3 — May 2024
Founder & Editor-in-Chief – UMass Index Yearbook — Aug. 2020 – Jul. 2022
- Revived printed yearbook: proposed concept, raised funds, and delivered ∼200 books.

– Seminars –

Grad Student Organizer – CS Systems & Sustainability Lunches, UMass Amherst — Fall 2024 – Spring 2025

Grad Student Organizer – CS Theory Seminar, UMass Amherst — Spring 2024 –

Grad Student Organizer – Algorithms with Predictions Seminar, UMass Amherst — Fall 2022

Open Source Projects

`citation-checker` – *command-line application, Python* — Spring 2026
Deterministic bibliography verifier for `.bib` files and PDFs that attempts to verify entries against databases such as DOI and CrossRef.

GlassTerm – *macOS application, Swift* — Spring 2026
My personal terminal app for macOS. Fast and lightweight with basic customization and tabs.

Venti – *macOS application, Swift & JavaScript* — Spring 2023, 2026
Carbon-aware battery management tool for Apple silicon MacBooks. Prolongs battery health and defers charging to periods of time when grid electricity is sufficiently clean.

Backtrack – *iOS application, Swift & SwiftUI* — Summer 2021
Privacy-centric app that provides location history; optimized for low power use using deep system APIs to reuse location fetches.

Béton3 macro pad & volume knob – *Hardware and firmware design, C, CAD, Arduino* — Summer 2020
Playback control input device based on an Arduino Pro Micro.

UMass Campus Architecture – *iOS, watchOS application, SwiftUI* — Summer 2020, Fall 2025
An interactive guide to highlight the architecture of the UMass campus.

Languages

English (U.S.) [native]; Polish [native]; American Sign Language [beginner]